OWNER : HINDUSTAN PETROLEUM CORPORATION LIMITED (HPCL)

PMC : ENGINEERS INDIA LIMITED, NEW DELHI (EIL)

UNIT : 504

PMC JOB NO. : B016

DOCUMENT SUBMISSION STATUS - LTHE	**REVIEW STATUS - EIL**
DOC. CATEGORY: (USE TICK MARK) ☐ R ☑ I	☐ **CODE 1 – NO COMMENTS. PROCEED WITH MANUFACTURE / FABRICATION / CONSTRUCTION AS PER THE DOCUMENT.** ☐ **CODE 2 – PROCEED WITH MANUFACTURE/FABRICATION/CONSTRUCTION AS PER COMMENTED DOCUMENT. REVISED DOCUMENT REQUIRED.** ☐ **REVIEW CODE 3 – DOCUMENT DOES NOT CONFORM TO BASIC REQUIREMENTS AS MARKED. RESUBMIT FOR REVIEW** ☐ **Code R – DOCUMENT IS RETAINED FOR RECORDS. PROCEED WITH MANUFACTURE / FABRICATION** ☐ **Code V - VOID**
ISSUED FOR	
☑ INFORMATION ☐ REVIEW ☐ CONSTRUCTION ☐ INQUIRY ☐ PROCUREMENT ☐ AS BUILT	NAME / DISCIPLINE / SIGN. / DATE

Revision No.	Description	Prepared by	Reviewed by	Approved by	Reviewed by	Approved by	Approved Date
02	Issued for Information	PPS	KDS	BDT	--	--	17-Mar-'20
01	Issued for Information	PPS	KDS	BDT	--	--	6-Feb-'20
00	Issued for Information	PPS	KDS	BDT	--	--	26-Nov-'19

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

REVISION HISTORY SHEET

REV. NO.	REV. DATE	REVISED PAGE	REVISION DESCRIPTION
00	26-Nov-'19	-	First Issue
01	6-Feb-'20	7	Updated document type, numbering system for process P&IDs, PFDs, & MSDs and series number for instrument process data sheets
02	17-Mar-'20	6, 8 and 10	Numbering System for Vendor Drawings / Documents

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

INDEX

1. SCOPE 4
2. RESPONSIBILITY 4
3. NUMBERING SYSTEM 4
4. REVISION CODING SYSTEM 10

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

1. SCOPE

This document decscribes the numbering system for Drawings / Documents / Requisitions / Vendor Drawing generated for the Residue Up-gradation Facility, Unit of Visakh Refinery Modernization Project at, Visakhapatnam.

2. RESPONSIBILITY

The Drawings / Documents / Requisitions number shall be allotted by the departments that originate them.

3. NUMBERING SYSTEM

3.1 Numbering system for L&T Drawings / Documents

B016-RUF-LT-504-XX-XX-0001
① ② ③ ④ ⑤ ⑥ ⑦

The Numbering shall include

1. EIL Job number i.e. **B016,** this number shall remain same for all documents covered under RUF project.
2. RUF – Residue Up-gradation Facility
3. EPCC Contractor name i.e. LT – L&T Hydrocarbon Engineering Limted, Vadodara
4. Unit Number as 504 : Residue Up-gradation Facility
5. Document initiating department
    - PE - Project Engineering
    - MS – Mechanical Engineering - Static
    - CV – Civil, Structural & Building
    - EL - Electrical
    - IN - Instrumentation & Control Systems
    - PP - Plant & Piping Engineering
    - MR - Machinery Engineering - Rotary
    - MP - Machinery Engineering - Packages

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

- PC - Process Engineering
- PM – Project Management
- QA : Quality Assurance
- TS : Technical Safety
- PI : Plant Integrity
- QC : Quality Control

6. Document Type
    - QD - Quality Documents / procedures
    - SP - Specification / Design Criteria
    - CH - Charts / Graphs
    - CE – Cause & Effect
    - DW - Drawing
    - FD - Fabrication drawing
    - DS - Data sheet
    - IS - Isometric Drawings (Line not falling under stress)
    - IC - Isometric Drawings (Line Falling under stress-Critical)
    - GA - General Arrangement Drawings
    - LS - List and Summary
    - MR - Material requisition
    - PR - Purchase requisition
    - TE - Technical Bid Evaluation
    - CL - Design Calculations
    - AR - Analysis reports
    - BQ - Bill of material
    - OM - Manuals
    - OD - Other documents
    - MS – Miscellaneous Document
    - GN – General
    - TR – Transmittal Note
    - CU – Curve

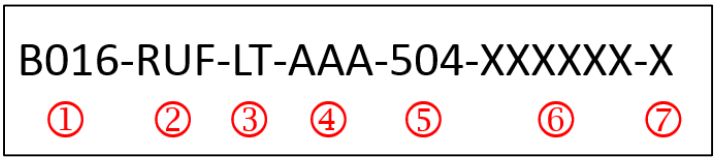
ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

- DI – Document Indices
- PD – Process Description

7. Progressive Number from 0001 to 9999,
   e.g. For equipment drawing it will be *B016-RUF-LT-504-MS-DW-0001*

3.2 Numbering system for Process P&IDs, PFD & MSDs



The number system shall include

1. EIL Job number i.e. **B016,** this number shall remain same for all documents covered under RUF project.

2. RUF – Residue Up-gradation Facility

3. EPCC Contractor name i.e. LT – L&T Hydrocarbon Engineering Limted, Vadodara

4. Deliverable type – AAA

   - PID – Piping & Instrumentation Diagram
   - PFD – Process Flow Diagram
   - MSD – Material selectin diagram

5. Unit Number as 504 : Residue Up-gradation Facility

6. Serial number

   e.g.: B016-RUF-LT-PID-504-11111

2

7. **Applicable only for Vendor P&IDs:** Additional P&ID's prepared by vendor shall be numbered as process P&ID containing the vendor supplied item with suffix A, B,-Z, as demonstrated below.

   Equipment vendor should number the P&ID from alphabet "A" to "Z". If serial numbers "A" to "Z" are not sufficient, vendor can use "A1" to "Z1" for subsequently.

   Examples:

   1. For 504-F-101, engineering P&ID number is B016-RUF-LT-PID-504-11117, hence vendor P&IDs will start with B016-RUF-LT-PID-504-11117-A

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

2. For 504-E-151 (Air preheater of 504-F-101) engineering P&ID number is B016-RUF-LT-504-11117B, hence vendor P&IDs will start with B016-RUF-LT-PID-504-11117B-A

3. For 504-K-101A engineering P&ID number is B016-RUF-LT-PID-504-11143, hence vendor P&IDs will start with B016-RUF-LT-PID-504-11143-A

4. For 504-P-101A/B engineering P&ID number is B016-RUF-LT-PID-504-11115-2, hence vendor P&IDs will start with B016-RUF-LT-PID-504-11115-2-A

3.3 Numbering system for Process Data sheets

B016-RUF-LT-504-PC-DS-XXYY
① ② ③ ④ ⑤ ⑥ ⑦ ⑧

The Numbering shall include

1. EIL Job number i.e. **B016,** this number shall remain same for all documents covered under RUF project.

2. RUF – Residue Up-gradation Facility

3. EPCC Contractor name i.e. LT – L&T Hydrocarbon Engineering Limted, Vadodara

4. Unit Number as 504 : Residue Up-gradation Facility

5. Document initiating department

   - PC - Process Engineering

6. Document Type

   - DS - Data sheet

7. Data sheet type indication by first two digits - XX

   - 01 – Columns
   - 02 , 03 – Vessels
   - 04 – Heat Exchangers
   - 05 – Air Coolers
   - 06 – Filters / Strainers
   - 07 – Compressors

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

- 08 – HPRT / PRE
- 09, 10 – Pumps
- 11 – Fired Heater
- 12 – Tanks
- 13 – Packages
- 14 – Miscellaneous
- 21 – Instrument data sheets

8. Progressive Number, from 01 to 99, for the last two digits - YY
   e.g. For process data sheet for column will be *B016-RUF-LT-504-PC-DS-0101*

3.4 Numbering System for Vendor Drawings / Documents

2

B016-RUF-LT-504-XX-XX-XXXX(Y)-AABBB

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

The number system shall include

1. EIL Job number i.e. **B016,** this number shall remain same for all documents covered under RUF project.
2. RUF – Residue Up-gradation Facility
3. EPCC Contractor name i.e. LT – L&T Hydrocarbon Engineering Limted, Vadodara
4. Unit Number as 504 : Residue Up-gradation Facility

2

5. LTHE Package / item owner discipline[1]

   - MS – Mechanical Engineering - Static
   - CV – Civil, Structural & Building
   - EL - Electrical
   - IN - Instrumentation & Control Systems
   - PP - Plant & Piping Engineering
   - MR - Machinery Engineering - Rotary

[1] Updated in Rev.02

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

- MP - Machinery Engineering - Packages
- PC - Process Engineering
- TS : Technical Safety
- PI : Plant Integrity [only applicable for HVAC vendor]
- QA : Quality Assurance
- QC : Quality Control

6. Document Type

- QD - Quality Documents / procedures
- SP - Specification / Design Criteria
- CH - Charts / Graphs
- DW - Drawing
- FD - Fabrication drawing
- DS - Data sheet
- IS - Isometric Drawings (Line not falling under stress)
- IC - Isometric Drawings (Line Falling under stress-Critical)
- GA - General Arrangement Drawings
- LS - List and Summary
- CL - Design Calculations
- AR - Analysis reports
- BQ - Bill of material
- MA - Manuals
- OD - Other documents
- GN – General
- CU – Curve

7. Vendor short Name (4 alphabets only), if a vendor get order for more than one item then for 1storder Y shall be 1, for 2nd order Y shall be 2 and so on so forth.
8. Document type indication by first two digit - AA

- 01 – Columns
- 02 , 03 – Vessels
- 04 – Heat Exchangers

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.

- 05 – Air Coolers
- 06 – Filters / Strainers
- 07 – Compressors
- 08 – HPRT / PRE
- 09, 10 – Pumps
- 11 – Fired Heater
- 12 – Tanks
- 13 – Packages
- 14 – Miscellaneous
- 21 – Instrument data sheets

2

9. 3 digit progressive number from 001 to 999, for the last three digits – BBB, however for items / package *where more than 1000 deliverables are envisaged may use 4* digit progressive numbers from 0001 to 9999.

   E.g. Fabrication Drawing of L&T-MFF on Single Order for Column

   B016-RUF-LT-504-MS-FD-LMFF (1)-01001

   Suppose LMFF (vendor short name) receives 3 orders for items such as reactors, columns and vessel, and then L&T-MFF to follow;

   LMFF (1) - for reactor, LMFF (2) - for columns, LMFF (3) - for vessels

   If vendor receives one order only, Vendor name to be followed by (1) i.e. LMFF(1)

For vendor P&ID refer section 3.2 point 7 numbering system for vendor P&IDs

4. REVISION CODING SYSTEM

Revision code for Drawings / documents shall commence with 00 and subsequent revisions shall be in incremental order of 1 for each revision (E.g. 01, 02, 03….).

Revision code for As-Built Drawings / Documents shall continue from previous revision number with revision description as “As Built”.

ALL RIGHTS RESERVED

THIS DOCUMENT AND ANY DATA AND INFORMATION CONTAINED THEREIN ARE CONFIDENTIAL AND THE PROPERTY OF L&T HYDROCARBON ENGINEERING (LTHE) AND THE COPYRIGHT THEREIN IS VESTED IN LTHE NO PART OF THIS DOCUMENT, DATA, OR INFORMATION SHALL BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER, EXCEPT WITH THE PRIOR WRITTEN PERMISSION OF LTHE.